Nishanth Ganapathy Palaniswami

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EDUCATION

New York University — New York City, NY
Master of Science in Computer Engineering · **GPA 3.965/4.0 · NYU Merit Scholarship** — May 2026
Coursework: Advanced Machine Learning · Big Data · Deep Learning · Computer Systems Architecture · Stern MBA Strategic Foresight
VIP Researcher: dynamic NYC mobility map using GIS + community surveys

PSG College of Technology — Coimbatore, India
Bachelor of Engineering in Computer Science and Engineering — June 2024

WORK EXPERIENCE

Got2eat, Runner-Up in Base44's Inaugural Accelerator Cohort — New York City, NY
AI Engineer — August 2026 – Present

- Built a **server-side LLM assistant** turning meal and transit requests into validated app filters with **fail-closed rate limits**
- Engineered **GTFS-derived subway + PATH routing** and MapLibre layers with door-to-door **one-transfer ETAs**
- Fixed overnight availability with **minute-level updates**; added reversible, **audit-logged repairs** and headless map checks
- Shipped React 19/TypeScript and Deno features for a community map of **meals under $15** on the web and iOS **(TestFlight)**
- Developed a **B2B partner dashboard** for listings, deals, and occupancy; launched a **loyalty-card pilot** from partner feedback

SAIL GTX Inc., backed by Cornell Tech's Runway Startup Studio — New York City, NY
Software Engineer Intern — September 2025 – May 2026

- Designed a deterministic A-to-B HTS revision engine (Python, SQL) **≥ 95% accuracy**, 1,000+ relevant changes per release
- Standardized duty rates across trade programs and **166 HTS releases (2018–2026)**, powering downstream duty calculation
- **Solely owned** the HTS duty-calculation pipeline (Python, SQLite, Parquet, FastAPI), versioned ingestion through duty math
- Drove **97% pipeline run reliability;** golden-test harnesses caught tariff-stacking regressions before customer releases
- Co-built, with the CTO, a production **HTS classification model** mapping product descriptions to tariff codes

Tagwebs Technologies — Coimbatore, India
Software Engineer Intern — April 2023 – May 2024

- Launched Edliy.com, an EdTech platform **(40+ STEM modules);** provisioned **500+ schools** with teacher and class accounts in 90 days
- Deployed **15+** Python/Django REST web apps on AWS EC2/S3 with CI/CD, **cutting build time by 25%; 99% uptime** via CloudWatch

PROJECTS AND PUBLICATIONS

Second Brain - Agentic Assistant over Gmail/Slack/Calendar/GitHub (Python, Claude Agent SDK, MCP) **(Code)**

- Enforced **19 deterministic guardrails** across 5 permission boundaries, CI-regression-tested after a prompt-injection bypass
- Devised a **hybrid local RAG** over **an Obsidian vault** - sqlite-vec + SQLite FTS5 with local ONNX embeddings (fastembed)
- Diff-gated scheduler invokes the LLM only on new signal; every outbound artifact lands in a drafts folder for review

Census Trade Warehouse - Orchestrated ELT (Dagster, dbt, DuckDB) **(Code)**

- Scheduled ELT over **monthly U.S. Census imports**, Dagster + dbt on DuckDB producing ranked partner and product-category marts
- **12 dbt data tests** + pytest in GitHub Actions CI; clones and runs offline via a bundled sample, no API key required

voicebot - Local Push-to-Talk Dictation (Python, faster-whisper, Ollama) **(Code)**

- Fully offline dictation - int8 Whisper STT, boot-time CUDA probe with CPU fallback, local LLM cleanup; **~2.5s on a 4 GB GPU**

MetroScan - NYC Subway Ridership and Delay Detection (PySpark, Spark MLlib) **(Code) (Slides)**

- Restructured a **110M-row** raw MTA feed (2020–24) into a 16M-row hourly feature table in PySpark - 2-person agile team

Parameter-Efficient Fine-Tuning - RoBERTa + LoRA on AG News (Code)

- **94.8% test accuracy** fine-tuning 0.80% of parameters (999K of 125.6M) with LoRA adapters - NYU DL, group project

American Sign Language Alphabet Recognition Using Teachable Machine - IEEE ICCCNT 2024 (Paper)

- Co-authored publication; 99% alphabet-set accuracy per the paper - DOI: 10.1109/ICCCNT61001.2024.10725877

SKILLS

Languages and Frontend: Python, SQL, TypeScript, JavaScript, React, Vite, Next.js, HTML/CSS
AI and ML: PyTorch, scikit-learn, pandas, Hugging Face Transformers, LoRA/PEFT, RAG, agents, MCP, Claude Agent SDK
Backend and Data: FastAPI, Django, Node.js, Deno, PostgreSQL, SQLite, PySpark, Dagster, dbt, DuckDB, Parquet, REST APIs
Geospatial and Mobile: MapLibre GL, GTFS, GeoJSON, Capacitor
Testing, CI and Cloud: pytest, golden tests, schema validation, Playwright, GitHub Actions, Git, AWS (EC2, S3, CloudWatch)